

Repeating Substring

Time limit: 1.00 s

Memory limit: 512 MB

A repeating substring is a substring that occurs in two (or more) locations in the string. Your task is to find the longest repeating substring in a given string.

Input

The only input line has a string of length n that consists of characters a–z.

Output

Print the longest repeating substring. If there are several possibilities, you can print any of them. If there is no repeating substring, print -1 .

Constraints

- $1 \leq n \leq 10^5$

Example

Input	Output
cabababc	abab